Fiscal Policy, Inflation Stabilization, and Monetary-Fiscal Interaction*

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Abstract

In a New Keynesian model in which fiscal policy directly affects aggregate demand and responds to both inflation and public debt, the standard active–passive fiscal boundary in Leeper (1991) is altered. This paper studies the conditions under which fiscal policy helps select a unique and stable equilibrium. The fiscal and monetary responses to inflation that result in a unique equilibrium depend on how strongly fiscal policy adjusts to stabilize debt. When the fiscal response to debt is weak, a stronger fiscal response to inflation can compensate for a weaker monetary response to inflation. When the fiscal response to debt is strong, a stronger fiscal response to inflation instead requires a stronger monetary response to inflation to maintain a unique and stable equilibrium. At the boundary value of the fiscal response to debt, equilibrium uniqueness is guaranteed by fiscal policy alone.

JEL codes: E31, E52, E62, H63

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1 Introduction

In the classic framework developed by Leeper (1991), fiscal policy is characterized as passive when an increase in government debt is followed by a sufficiently strong increase in future primary surpluses. Monetary policy is passive when the nominal interest rate does not rise strongly enough in response to inflation. Equilibrium is stable and unique when one authority is active and the other is passive. The fiscal and monetary active–passive regions are defined independently of each other: the fiscal policy characterization depends on the response of the primary surplus to debt, while the monetary policy characterization depends on the response of the nominal interest rate to inflation. In Leeper (1991), fiscal policy matters for uniqueness through debt adjustment rather than through a direct effect on aggregate demand.

I use a parsimonious New Keynesian model with two fiscal ingredients added to the textbook three-equation system. First, the primary surplus enters the Euler equation, so fiscal policy directly shifts aggregate demand. Second, the fiscal authority follows a rule under which the primary surplus responds to current inflation, in addition to debt.

Once fiscal policy also affects aggregate demand directly, the active–passive regions in Leeper (1991) are no longer sufficient to guide determinacy analysis. The fiscal boundary between active and passive fiscal policy is altered, and the interaction between fiscal and monetary responses to inflation depends on how strongly fiscal policy responds to debt. Under weak fiscal backing, a strong fiscal response to inflation can compensate for a weak monetary response to inflation, and vice versa. Under strong fiscal backing, the logic reverses: a stronger fiscal response to inflation requires a stronger monetary response to inflation to obtain uniqueness. There is a value of the fiscal response to debt where local uniqueness is pinned down by the fiscal response to inflation, while the monetary policy response to inflation has no role in determining uniqueness.

These results contribute to several strands of the literature. One examines environments in which fiscal policy affects private spending directly, including models with bond-in-utility, finite horizons, and household heterogeneity (Blanchard, 1985; Kaplan et al., 2018; Michaillat and Saez, 2021). In these environments, my paper shows how the active–passive fiscal threshold in Leeper (1991) and Cochrane (2023) changes once fiscal policy enters the demand block directly. I also contribute to work on fiscal feedback rules as stabilization tools, including fiscal rules that respond to inflation (Cochrane, 2023). While that literature emphasizes how fiscal adjustment helps stabilize debt, inflation, and the price level (Kirsanova and Wren-Lewis, 2012; Schmitt-Grohé and Uribe, 2000; Woodford, 2001), I show that fiscal policy can affect equilibrium uniqueness not only through debt adjustment, but also through its direct effect on aggregate demand and its interaction with monetary policy.

The remainder of the paper is organized as follows. Section 2 presents the model. Section 3 characterizes the determinacy regions under alternative fiscal responses. Section 4 extends the baseline analysis. Section 5 concludes.

2 Model

I study a parsimonious New Keynesian model, building on Cochrane (2023), to examine how fiscal policy affects equilibrium determinacy when the primary surplus both enters aggregate demand and responds to inflation. Relative to the textbook three-equation New Keynesian system, I introduce two fiscal ingredients. First, the primary surplus enters the Euler equation, so fiscal policy affects private demand directly. Second, government debt evolves according to a budget constraint, and the fiscal authority sets the primary surplus as a function of current inflation and lagged debt.

The model is given by

$$x_t = E_t x_{t+1} - \sigma \left(i_t - E_t \pi_{t+1} \right) + \gamma s_t, \tag{1}$$

$$\pi_t = \beta E_t \pi_{t+1} + \kappa x_t, \tag{2}$$

$$\rho v_t = v_{t-1} + i_{t-1} - \pi_t - s_t, \tag{3}$$

$$s_t = \psi \pi_t + \alpha v_{t-1}, \tag{4}$$

$$i_t = \theta \pi_t. \tag{5}$$

The endogenous variables are the output gap x_t, inflation π_t, the nominal interest rate i_t, the primary surplus s_t, and real government debt v_t. Expectations are rational and conditional on information available at time t.

Equation (1) is the Euler equation. Current demand depends on expected future demand, the ex ante real interest rate, and the primary surplus. The parameter $\sigma > 0$ governs the sensitivity of demand to the real interest rate. The parameter γ governs the effect of the primary surplus on private demand. Throughout the paper, I assume $\gamma < 0$, so a higher primary surplus lowers demand, while a larger deficit raises demand.

Equation (2) is the standard New Keynesian Phillips curve. Current inflation depends on expected future inflation and the output gap. The parameter $\beta \in (0, 1)$ is the discount factor, and $\kappa > 0$ measures how strongly real activity feeds into inflation.

Equation (3) is the law of motion for (log) real government debt. Debt today depends on lagged debt, lagged nominal interest payments, current inflation, and the primary surplus. The parameter $\rho \in (0, 1)$ is a linearization constant inherited from the government budget constraint.*

Equation (4) specifies fiscal policy. The primary surplus responds to contemporaneous inflation and to lagged debt. The parameter ψ measures the fiscal response to inflation, while the parameter α measures the fiscal response to debt. Thus, ψ governs how strongly fiscal policy tightens or loosens in response to inflation, and α governs the strength of debt feedback.

Equation (5) specifies monetary policy. The nominal interest rate responds to inflation with coefficient θ, so θ measures the strength of the monetary inflation response. This

*See Appendix A for the derivation. Starting from the nominal government budget constraint, I divide by the price level to express variables in real terms and then log-linearize around the steady state. This yields $\rho v_t = v_{t-1} + i_{t-1} - \pi_t - s_t$. In this representation, ρ collects steady-state terms in the linearized debt equation; see Cochrane (2023).

parameter has typically played a key role in the literature that examines the stability and uniqueness of equilibrium.

The model has two forward-looking variables, the output gap x_t and inflation π_t. Equilibrium determinacy therefore depends on whether expectations pin down a unique path for these two variables. The relevant criterion is the Blanchard–Kahn condition. In a linear rational-expectations model, local uniqueness requires that the number of unstable roots match the number of forward-looking variables. Here that means exactly two eigenvalues must lie outside the unit circle.

Substituting the surplus rule and the Taylor rule into the system, and then using the Phillips curve to eliminate x_t, reduces the model to two dynamic equations in inflation and debt. Consider a homogeneous solution $\pi_t = \Pi\lambda^t$ and $v_t = V\lambda^t$. Using $E_t\pi_{t+1} = \lambda\pi_t$, $E_t\pi_{t+2} = \lambda^2\pi_t$, $\pi_{t-1} = \lambda^{-1}\pi_t$, and $v_{t-1} = \lambda^{-1}v_t$, the system yields the quartic characteristic equation

$$P(\lambda;\theta,\psi,\alpha) \equiv c_4\lambda^4 + c_3\lambda^3 + c_2\lambda^2 + c_1\lambda + c_0, \tag{6}$$

where the coefficients are given by

$$\begin{aligned}
c_4 &= \rho\beta, \\
c_3 &= -\rho(1+\beta+\kappa\sigma) + (\alpha-1)\beta, \\
c_2 &= \rho + \rho\kappa\sigma\theta - \rho\kappa\gamma\psi - (\alpha-1)(1+\beta+\kappa\sigma), \\
c_1 &= (\alpha-1)\big(1+\kappa\sigma\theta-\kappa\gamma\psi\big) + \kappa\gamma\alpha(1+\psi), \\
c_0 &= -\kappa\gamma\alpha\theta.
\end{aligned} \tag{7}$$

A nonzero solution $(\Pi, V) \neq (0,0)$ therefore requires $P(\lambda;\theta,\psi,\alpha) = 0$.

3 Results

I begin with Leeper (1991). In that framework, equilibrium determinacy depends on the monetary response to inflation, θ, and the fiscal response to debt, α. In Cochrane (2023),[†] $\alpha > 1-\rho$ corresponds to the standard specification of passive fiscal policy. The key departure from both Leeper (1991) and Cochrane (2023) is that fiscal policy affects equilibrium through two additional channels. The primary surplus enters aggregate demand directly through γ, and the fiscal rule also responds to inflation through ψ. As a result, the policy map changes.

Figure 1 plots the case with $\psi = 0$. The (α, θ) plane still resembles a Leeper-style policy map, but the vertical fiscal boundary is no longer given by $1-\rho$. Instead, it is given by

$$\bar{\alpha} = \frac{\sigma(1-\rho)}{\sigma - \gamma}.$$

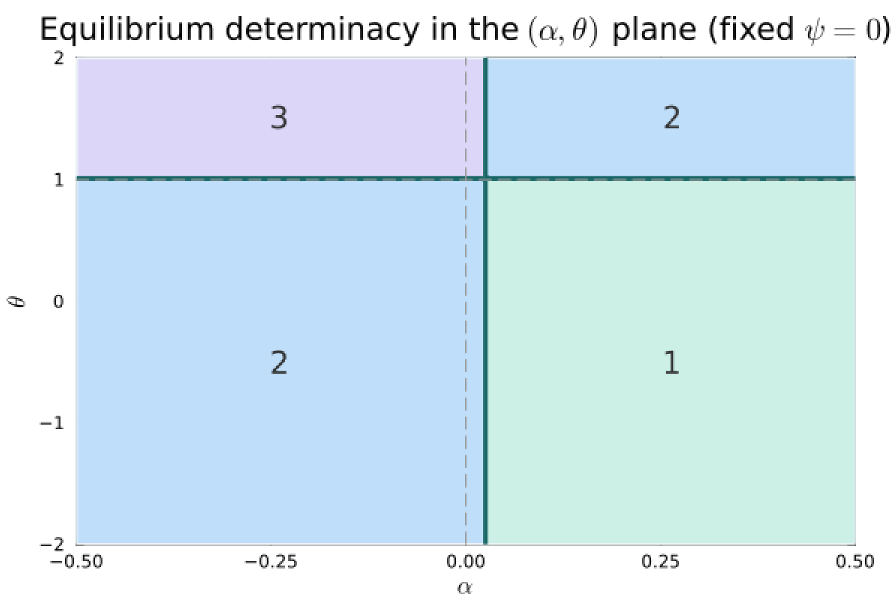

Figure 1: Equilibrium determinacy in the (α, θ) plane with $\psi = 0$ when fiscal policy affects demand ($\gamma \neq 0$). Numbers indicate the count of eigenvalues with modulus greater than one. The vertical boundary in α is the active–passive fiscal boundary $\alpha = \bar{\alpha} = \sigma(1-\rho)/(\sigma-\gamma)$. Parameters: $\kappa = 0.5$, $\beta = 0.99$, $\rho = 0.95$, $\sigma = 0.5$, and $\gamma = -0.5$.

If $\alpha > \bar{\alpha}$, debt feedback is sufficiently strong, so fiscal policy is passive. If $\alpha < \bar{\alpha}$, debt

[†]See Cochrane (2023), pp. 102 and 138.

feedback is not sufficiently strong, so fiscal policy is active. The subsection below derives this boundary. Appendix B provides the underlying root analysis.

3.1 The Boundary Value of the Fiscal Response to Debt

A natural place to begin is at the point where one of the eigenvalues reaches one, because this is where the Blanchard–Kahn root count can change. This means $\lambda = 1$ in equation 6. To isolate the fiscal boundary for the response to debt, first set $\psi = 0$, so fiscal policy does not respond to inflation. Evaluating the characteristic polynomial at $\lambda = 1$ then gives

$$P(1;\theta,0,\alpha) = \kappa\theta\big[(\sigma-\gamma)\alpha - \sigma(1-\rho)\big].$$

For any given $\kappa, \theta \neq 0$ and $\sigma \neq \gamma$, the condition $P(1;\theta,0,\alpha) = 0$ is therefore equivalent to

$$\alpha = \frac{\sigma(1-\rho)}{\sigma-\gamma}.$$

I define the fiscal active-passive threshold ($\bar{\alpha}$) as

$$\bar{\alpha} \equiv \frac{\sigma(1-\rho)}{\sigma-\gamma}. \tag{8}$$

It is the value of the debt-response coefficient at which a root reaches $\lambda = 1$, so it marks the point where the Blanchard–Kahn root count can change as fiscal policy moves from one region to the other. In particular, in Leeper's terminology, $\alpha < \bar{\alpha}$ implies active fiscal policy, whereas $\alpha > \bar{\alpha}$ implies passive fiscal policy.

More generally, evaluating the characteristic polynomial at $\lambda = 1$ gives

$$P(1;\theta,\psi,\alpha) = \kappa\Big[(\sigma-\gamma)(\alpha-\bar{\alpha})\theta + (1-\rho)\gamma\psi\Big]. \tag{9}$$

When $\gamma = 0$, so that the surplus does not enter the Euler equation directly, the fiscal boundary reduces to $\bar{\alpha} = 1 - \rho$, as in Cochrane (2023). For the case where $\gamma < 0$,

$$\bar{\alpha} < 1 - \rho.$$

Once the primary surplus also affects current demand, fiscal policy works through an additional channel. A higher surplus now restrains spending immediately, not only through future debt stabilization. Because part of the adjustment occurs on impact, a smaller debt response is enough to reach the fiscal threshold. That is why the threshold shifts downward relative to the benchmark classification in Leeper (1991) and the fiscal threshold in Cochrane (2023).

I now turn to the full model with $\psi \neq 0$, where the primary surplus also responds to inflation. The interaction between the fiscal response to inflation and the monetary response to inflation depends on whether α lies below, at, or above $\bar{\alpha}$.

3.2 Fiscal and monetary responses to inflation

Given the fiscal response to debt, I now examine how the fiscal and monetary responses to inflation interact to determine equilibrium uniqueness.

3.2.1 Boundary case: $\alpha = \bar{\alpha}$

Setting $\alpha = \bar{\alpha}$ in (9) gives

$$P(1; \theta, \psi, \bar{\alpha}) = \kappa(1 - \rho)\gamma\psi. \tag{10}$$

At this boundary, the monetary coefficient θ drops out of the unit-root condition. The fiscal boundary for uniqueness is $\psi = 0$. When $\psi < 0$, the equilibrium is unique. This is shown in Appendix B.3.

When $\alpha = \bar{\alpha}$, local uniqueness is pinned down by the fiscal response to inflation, not by the Taylor-rule coefficient. Figure 2 shows this. The boundary between uniqueness and multiplicity is vertical in the (ψ, θ) plane: moving left or right in ψ changes the root count,

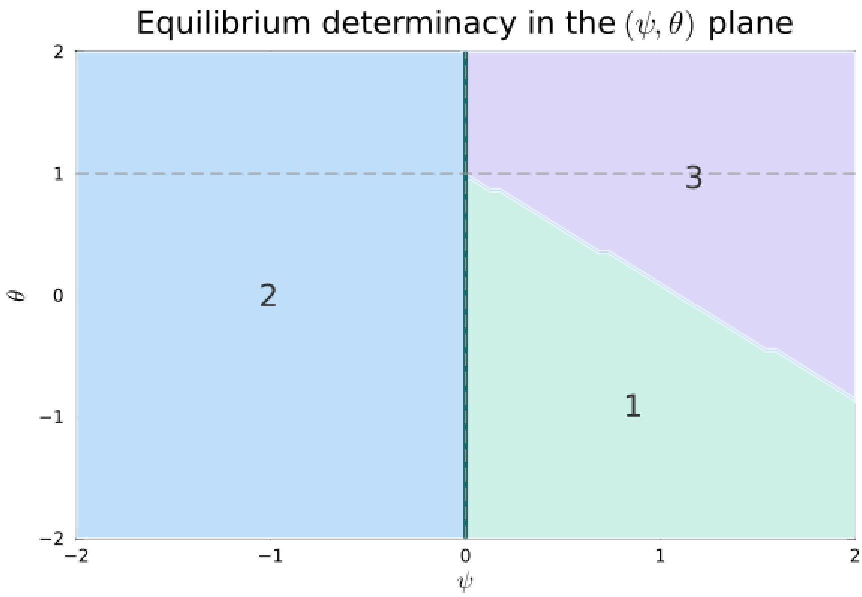

Figure 2: Equilibrium determinacy in the (ψ, θ) plane for the boundary case $\alpha = \bar{\alpha} = 0.025$. Numbers indicate the count of eigenvalues outside the unit circle. With two forward-looking variables, local uniqueness requires exactly two unstable roots (region "2"). Parameters: $\kappa = 0.5$, $\beta = 0.99$, $\rho = 0.95$, $\sigma = 0.5$, and $\gamma = -0.5$.

while moving up or down in θ does not.

This case has no counterpart in the policy map of Leeper (1991): when the debt-response coefficient is $\alpha = \bar{\alpha}$, fiscal policy alone determines whether equilibrium is unique.

3.2.2 Weak fiscal backing: $\alpha < \bar{\alpha}$

When $\alpha < \bar{\alpha}$, debt feedback is weak. In this region, the fiscal response to inflation mainly works through current demand. The fiscal and monetary responses to inflation therefore act as close substitutes in supporting equilibrium uniqueness. Figure 3 illustrates the case.

The boundary in the (ψ, θ) plane slopes downward. When the fiscal authority raises the primary surplus more strongly in response to inflation, a smaller monetary response is enough to obtain uniqueness. Both policies lean against inflation through the same current-demand channel. A higher θ lowers demand through the real interest rate. A higher ψ lowers demand because a higher surplus directly reduces spending in the Euler equation.

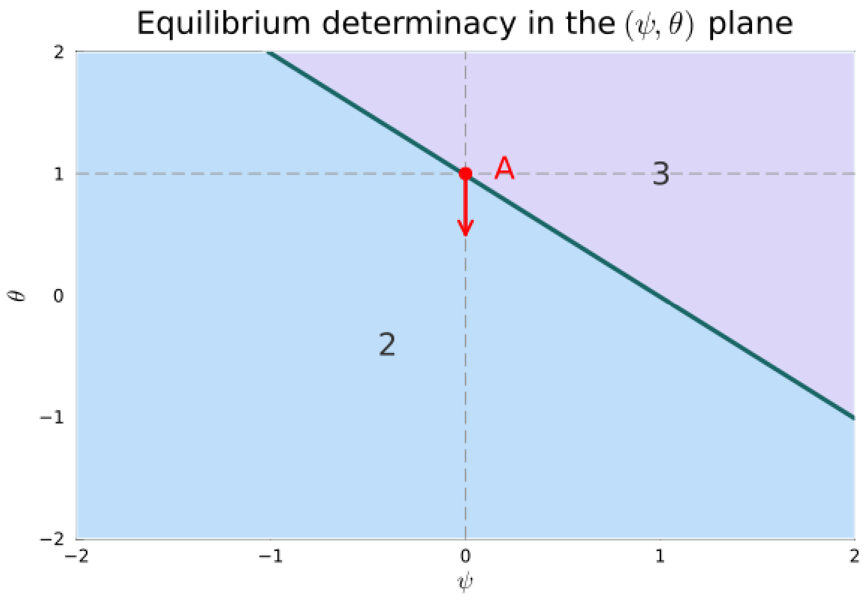

Figure 3: Equilibrium determinacy in the (ψ, θ) plane for $\alpha = 0$. Numbers indicate the count of eigenvalues with modulus greater than one. With two forward-looking variables, local uniqueness requires exactly two unstable roots (region "2"). Parameters: $\kappa = 0.5$, $\beta = 0.99$, $\rho = 0.95$, $\sigma = 0.5$, and $\gamma = -0.5$.

In this special case $\alpha = 0$, the characteristic roots can be written in closed form:

$$\lambda_1 = 0, \quad \lambda_2 = 1 + \frac{1 + \kappa\sigma - \beta - \sqrt{(1 + \kappa\sigma - \beta)^2 + 4\beta\kappa\left(-\sigma\theta + \gamma\psi + \sigma\right)}}{2\beta},$$

$$\lambda_3 = \frac{1}{\rho}, \quad \lambda_4 = 1 + \frac{1 + \kappa\sigma - \beta + \sqrt{(1 + \kappa\sigma - \beta)^2 + 4\beta\kappa\left(-\sigma\theta + \gamma\psi + \sigma\right)}}{2\beta}.$$

Since $\lambda_1 = 0$, $\lambda_3 = 1/\rho > 1$, and $\lambda_4 > 1$, uniqueness requires $\lambda_2 < 1$, which is equivalent to

$$\sigma(1 - \theta) + \gamma\psi > 0 \quad \Longleftrightarrow \quad \theta - \frac{\gamma}{\sigma}\psi < 1.$$

So when $\alpha = 0$, the determinacy boundary is a straight line, as in Figure 3. The point A at $(\psi, \theta) = (0, 1)$ is the standard Leeper benchmark. Starting from A, movements along the boundary into region "2" describe policy combinations that preserve local uniqueness. In particular, the downward direction from A corresponds to the unique and stable passive-

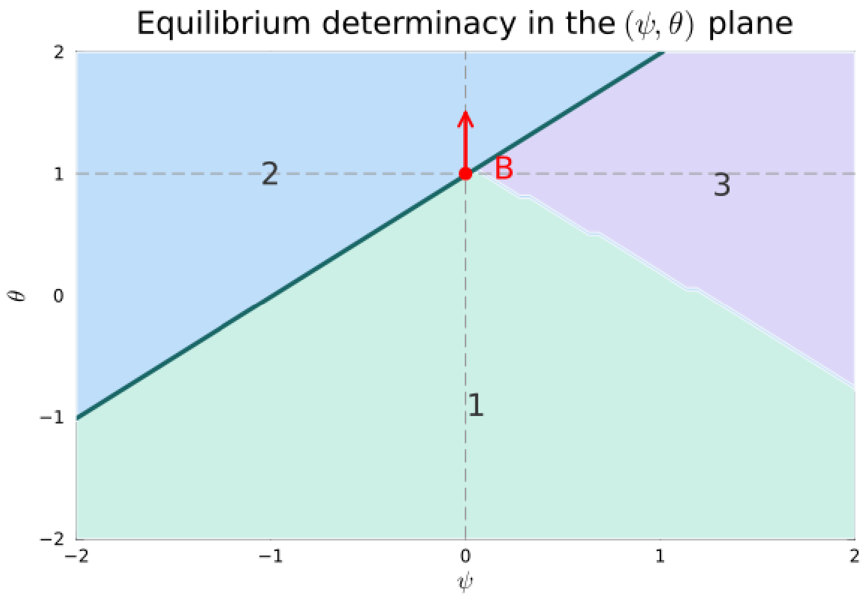

Figure 4: Equilibrium determinacy in the (ψ, θ) plane for $\alpha = 0.05$. Numbers indicate the count of eigenvalues with modulus greater than one. With two forward-looking variables, local uniqueness requires exactly two unstable roots (region "2"). Parameters: $\kappa = 0.5$, $\beta = 0.99$, $\rho = 0.95$, $\sigma = 0.5$, and $\gamma = -0.5$.

monetary, active-fiscal region in Leeper (1991). The figure also shows that the unique region extends beyond the standard Leeper case once the primary surplus enters the Euler equation directly. By contrast, if monetary and fiscal policy both respond too aggressively to inflation, the economy moves into the explosive active–active region.

This region also connects to the insight in Leeper (2009) and the "stepping on the rake" literature (Cochrane, 2023, 2018; Sims, 2011) that a stronger monetary response alone does not always guarantee determinacy when fiscal backing is weak. In the present model, when $\alpha < \bar{\alpha}$, increasing θ can push an additional root outside the unit circle and destroy uniqueness.

3.2.3 Strong fiscal backing: $\alpha > \bar{\alpha}$

When $\alpha > \bar{\alpha}$, debt feedback is strong. In this region, a stronger fiscal response to inflation makes uniqueness harder to obtain: the monetary response required for uniqueness rises with ψ. Figure 4 shows this pattern for $\alpha = 0.05$.

A higher ψ raises the primary surplus on impact, which lowers demand today. But it also lowers real debt through the government budget constraint. Because the fiscal rule responds to lagged debt, lower debt today implies smaller debt-driven primary surpluses tomorrow. In other words, a stronger fiscal response to inflation tightens today but weakens future fiscal restraint.

Under $\gamma < 0$, a smaller expected path of future primary surpluses supports higher expected demand. Through the Phillips curve, that puts upward pressure on expected inflation. In the strong fiscal-backing region, a stronger fiscal response to inflation therefore makes inflation expectations harder to anchor, not easier. A stronger monetary response is needed to preserve uniqueness, which is why the boundary in Figure 4 slopes upward.

Figure 4 also shows that uniqueness is not confined to the point B at $(\psi, \theta) = (0, 1)$, which corresponds to the standard Leeper benchmark. The upward direction from C enters region "2" and recovers the active-monetary, passive-fiscal region in Leeper (1991). Once the primary surplus enters the Euler equation, however, the unique region extends beyond that benchmark case. In particular, when $\psi < 0$, fiscal policy lowers the primary surplus when inflation rises, and the economy can still satisfy the Blanchard–Kahn condition even with a Taylor-rule coefficient below one.

4 Extension

I now consider an alternative way in which fiscal policy can affect aggregate demand. Replace the Euler equation in the baseline model with

$$x_t = E_t x_{t+1} - \sigma\big(i_t - E_t \pi_{t+1}\big) + \chi v_t, \tag{11}$$

while leaving the New Keynesian Phillips curve, the government budget constraint, the surplus rule, and the Taylor rule unchanged; that is, Equations (2)–(5) continue to hold.

The key difference from the baseline model is that demand now depends on the level of government debt, v_t, rather than on the primary surplus, s_t. This specification is in the spirit of bond-in-utility models, where holding government liabilities provides utility; see, for example, Kaplan (2025).

The main point of the extension is that the qualitative logic of the baseline model survives this change. There is again a boundary value for the debt-response coefficient,

$$\bar{\alpha}_v \equiv 1 - \rho - \frac{\chi}{\sigma}, \tag{12}$$

which organizes how the fiscal response to inflation interacts with the monetary response to inflation. Appendix C derives this boundary.

The economic interpretation is parallel to the baseline model. Once a fiscal variable enters the demand block, the relevant fiscal boundary is no longer determined by debt dynamics alone. In the baseline model, the key additional channel came from the primary surplus entering the Euler equation. Here it comes from government debt entering the Euler equation. The specific state variable is different, but the logic is the same: fiscal policy now affects determinacy through demand as well as through debt adjustment.

The boundary case $\alpha = \bar{\alpha}_v$ delivers the same result as in the baseline model. At this value, local uniqueness is pinned down by the fiscal response to inflation, while the Taylor-rule inflation coefficient does not affect the determinacy boundary. Monetary policy still matters for equilibrium dynamics, but it does not determine the boundary between uniqueness and multiplicity.

When $\alpha < \bar{\alpha}_v$, debt feedback is weak. In this region, the fiscal response to inflation behaves more like a same-period demand-control tool. The implication again matches the baseline model: the fiscal and monetary responses to inflation act as close substitutes for equilibrium uniqueness. A stronger fiscal response to inflation is therefore associated with a smaller monetary response required for determinacy, and the boundary in the (ψ, θ) plane slopes downward.

When $\alpha > \bar{\alpha}_v$, debt feedback is strong. In this region, a stronger fiscal response to inflation makes uniqueness harder to obtain, so the monetary response required for determinacy rises with ψ. The determinacy boundary in the (ψ, θ) plane therefore slopes upward, just as in the baseline model. When fiscal policy responds more strongly to inflation, it changes the expected path of future fiscal adjustment through its effect on debt, and this makes inflation expectations harder to anchor unless monetary policy also responds more aggressively.

Figures 5–7 in Appendix C show these three cases. When $\alpha > \bar{\alpha}_v$, the boundary in the (ψ, θ) plane slopes upward. When $\alpha < \bar{\alpha}_v$, it slopes downward. When $\alpha = \bar{\alpha}_v$, it is vertical in ψ. The same three-region structure therefore reappears in this alternative specification.

This extension strengthens the main interpretation of the paper. The central result does not depend on whether the primary surplus enters the Euler equation directly. More generally, once fiscal variables directly affect aggregate demand, a fiscal boundary emerges that organizes the interaction between fiscal and monetary responses to inflation. In that environment, the Taylor-rule inflation coefficient alone does not anchor uniqueness.

5 Conclusion

This paper studies how fiscal policy changes equilibrium determinacy when it affects aggregate demand directly. In the model, the primary surplus influences private spending and also responds to inflation and debt. Once fiscal policy enters the demand block in this way, the standard mapping from fiscal and monetary rules to equilibrium uniqueness changes relative to Leeper (1991).

The first main result is that the fiscal active–passive boundary in Leeper (1991) is altered. In that framework, the relevant fiscal boundary is pinned down by debt stabilization alone. Here, it also depends on how strongly fiscal policy affects current demand. This changes the classification of fiscal regimes for determinacy analysis.

The second main result is that this altered fiscal boundary also governs how the fiscal response to inflation interacts with the monetary response to inflation. When debt feedback is weak, the fiscal and monetary responses to inflation act as close substitutes: a weaker monetary response can be compensated by a stronger fiscal response. When debt feedback is strong, a stronger fiscal response to inflation instead requires a stronger monetary response to maintain uniqueness. At the boundary itself, local uniqueness is pinned down by the fiscal response to inflation, while the Taylor-rule coefficient drops out of the determinacy condition.

These results change the usual reading of monetary–fiscal interactions. In Leeper (1991), fiscal policy matters for determinacy through debt adjustment. In the present setting, fiscal policy also matters through its effect on current demand. As a result, the determinacy properties of the economy depend not only on how fiscal policy stabilizes debt, but also on how fiscal and monetary policy respond to inflation jointly.

The broader implication is that determinacy analysis should be reconsidered in modern macroeconomic environments in which fiscal variables affect private spending directly, including heterogeneous-agent models and other settings that depart from the representative-agent benchmark. More generally, once fiscal policy enters the demand block, debt backing alone is no longer enough to characterize equilibrium determinacy.

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A Derivation of the debt dynamics equation

This appendix derives Equation 3 and clarifies the origin of the coefficient ρ. Start from the nominal government budget constraint,

$$B_t = R_{t-1} B_{t-1} - P_t S_t, \tag{13}$$

where B_t is nominal debt at the end of period t, $R_{t-1} \equiv 1 + i_{t-1}$ is the gross nominal interest rate from $t-1$ to t, P_t is the price level, and S_t is the real primary surplus.

Divide Equation 13 by P_t to obtain the real constraint,

$$b_t \equiv \frac{B_t}{P_t} = R_{t-1}\frac{B_{t-1}}{P_t} - S_t = R_{t-1}\frac{B_{t-1}}{P_{t-1}}\frac{P_{t-1}}{P_t} - S_t, \tag{14}$$

and define gross inflation $\Pi_t \equiv P_t/P_{t-1}$. Then $P_{t-1}/P_t = \Pi_t^{-1}$, so

$$b_t = R_{t-1}\Pi_t^{-1}b_{t-1} - S_t. \tag{15}$$

To obtain the log-linear form used in Cochrane (2023), work with the debt-to-output ratio. Let real output be y_t, and define

$$v_t \equiv \log\left(\frac{b_t}{y_t}\right), \qquad g_t \equiv \log\left(\frac{y_t}{y_{t-1}}\right). \tag{16}$$

Define the (log) gross nominal return $r_t^n \equiv \log R_t$ and (log) gross inflation $\pi_t \equiv \log \Pi_t$, so the (log) ex post real return satisfies

$$r_t = r_{t-1}^n - \pi_t. \tag{17}$$

A first-order Taylor expansion of the exact nonlinear flow identity around a steady state with constant $\bar{r}$ and $\bar{g}$ yields the linearized debt-evolution equation

$$\rho\, v_t = v_{t-1} + r_t - g_t - \widetilde{s}_t. \tag{18}$$

Here $\widetilde{s}_t$ is the surplus-to-output ratio scaled by the steady-state debt-to-output ratio (so that it enters in the same units as v_t). The coefficient $\rho \in (0,1)$ is a constant of linearization. It depends only on steady-state terms (in particular on $\bar{r}$ and $\bar{g}$) and is introduced in Cochrane (2023) to express the flow equation in a convenient present-value form.

Finally, using $r_t = r^n_{t-1} - \pi_t$ and writing $i_{t-1} \equiv r^n_{t-1}$, Equation 18 becomes

$$\rho\, v_t = v_{t-1} + i_{t-1} - \pi_t - s_t, \tag{19}$$

which is Equation 3 in the main text under the normalization $s_t \equiv g_t + \widetilde{s}_t$ (i.e., absorbing the output-growth term into the surplus term).

B Characteristic roots and policy sensitivities in the baseline model

This appendix reports the root analysis behind the results in Section 3. The baseline model is

$$x_t = E_t x_{t+1} - \sigma\big(i_t - E_t \pi_{t+1}\big) + \gamma s_t, \tag{20}$$

$$\pi_t = \beta E_t \pi_{t+1} + \kappa x_t, \tag{21}$$

$$\rho v_t = v_{t-1} + i_{t-1} - \pi_t - s_t, \tag{22}$$

$$s_t = \psi \pi_t + \alpha v_{t-1}, \tag{23}$$

$$i_t = \theta \pi_t. \tag{24}$$

Throughout, $\kappa > 0$, $\sigma > 0$, $\rho \in (0, 1)$, and $\gamma < 0$.

B.1 The quartic characteristic polynomial

Substituting the fiscal and monetary policy rules into the baseline system gives

$$x_t = E_t x_{t+1} - \sigma\theta\pi_t + \sigma E_t \pi_{t+1} + \gamma\psi\pi_t + \gamma\alpha v_{t-1}, \tag{25}$$

$$\rho v_t = (1 - \alpha) v_{t-1} + \theta \pi_{t-1} - (1 + \psi)\pi_t. \tag{26}$$

Using the Phillips curve and substituting these expressions into (25), we obtain

$$\beta E_t \pi_{t+2} - (1 + \beta + \kappa\sigma)\, E_t \pi_{t+1} + (1 + \kappa\sigma\theta - \kappa\gamma\psi)\, \pi_t - \kappa\gamma\alpha v_{t-1} = 0. \tag{27}$$

The model is therefore reduced to the two equations (27) and (26) in inflation and debt.

Now consider a homogeneous solution

$$\pi_t = \Pi\lambda^t, \qquad v_t = V\lambda^t.$$

Then

$$E_t\pi_{t+1} = \lambda\pi_t, \qquad E_t\pi_{t+2} = \lambda^2\pi_t, \qquad \pi_{t-1} = \lambda^{-1}\pi_t, \qquad v_{t-1} = \lambda^{-1}v_t.$$

Substituting these into (27) and (26) gives

$$\left[\beta\lambda^3 - (1 + \beta + \kappa\sigma)\, \lambda^2 + (1 + \kappa\sigma\theta - \kappa\gamma\psi)\, \lambda\right] \Pi - \kappa\gamma\alpha V = 0, \tag{28}$$

$$\left[\theta - (1 + \psi)\lambda\right] \Pi + \left[(1 - \alpha) - \rho\lambda\right] V = 0. \tag{29}$$

A nontrivial solution $(\Pi, V) \neq (0, 0)$ requires the determinant of this 2×2 system to be zero:

$$\det \begin{pmatrix} \beta\lambda^3 - (1 + \beta + \kappa\sigma)\lambda^2 + (1 + \kappa\sigma\theta - \kappa\gamma\psi)\lambda & -\kappa\gamma\alpha \\ \theta - (1 + \psi)\lambda & (1 - \alpha) - \rho\lambda \end{pmatrix} = 0. \tag{30}$$

Expanding the determinant yields the quartic characteristic equation

$$P(\lambda; \theta, \psi, \alpha) \equiv c_4\lambda^4 + c_3\lambda^3 + c_2\lambda^2 + c_1\lambda + c_0 = 0, \tag{31}$$

with coefficients

$$c_4 = \rho\beta,$$

$$c_3 = -\rho(1 + \beta + \kappa\sigma) + (\alpha - 1)\beta,$$

$$
\begin{aligned}
c_2 &= \rho + \rho\kappa\sigma\theta - \rho\kappa\gamma\psi - (\alpha - 1)(1 + \beta + \kappa\sigma), \\
c_1 &= (\alpha - 1)\big(1 + \kappa\sigma\theta - \kappa\gamma\psi\big) + \kappa\gamma\alpha(1 + \psi), \\
c_0 &= -\kappa\gamma\alpha\theta.
\end{aligned} \tag{32}
$$

B.2 Implicit-function derivatives for a simple root

Let $\lambda(\theta, \psi, \alpha)$ be a simple root of $P(\lambda; \theta, \psi, \alpha) = 0$, i.e., $P(\lambda; \theta, \psi, \alpha) = 0$ and $P_\lambda(\lambda; \theta, \psi, \alpha) \neq 0$. Then

$$
\frac{\partial \lambda}{\partial \theta} = -\frac{P_\theta(\lambda; \theta, \psi, \alpha)}{P_\lambda(\lambda; \theta, \psi, \alpha)}, \qquad \frac{\partial \lambda}{\partial \psi} = -\frac{P_\psi(\lambda; \theta, \psi, \alpha)}{P_\lambda(\lambda; \theta, \psi, \alpha)}. \tag{33}
$$

Because θ enters only (c_2, c_1, c_0) and ψ enters only (c_2, c_1), differentiation of (32) gives

$$
P_\theta(\lambda; \theta, \psi, \alpha) = \kappa\sigma\big(\rho\lambda^2 + (\alpha - 1)\lambda\big) - \kappa\gamma\alpha, \tag{34}
$$

$$
P_\psi(\lambda; \theta, \psi, \alpha) = \kappa\gamma\, \lambda\, (1 - \rho\lambda), \tag{35}
$$

$$
P_\lambda(\lambda; \theta, \psi, \alpha) = 4c_4\lambda^3 + 3c_3\lambda^2 + 2c_2\lambda + c_1. \tag{36}
$$

B.3 Unit-root accounting and the $\bar{\alpha}$ threshold

A key object in the main text is the threshold

$$
\bar{\alpha} \equiv \frac{\sigma(1 - \rho)}{\sigma - \gamma}.
$$

This threshold appears transparently when we evaluate the characteristic polynomial at $\lambda = 1$.

Evaluate $P(1)$. Using (31)–(32),

$$
P(1; \theta, \psi, \alpha) = \sum_{j=0}^{4} c_j = \kappa\Big[(\sigma - \gamma)(\alpha - \bar{\alpha})\theta + (1 - \rho)\gamma\psi\Big]. \tag{37}
$$

The factorization in (37) is the central algebraic reason why $\bar{\alpha}$ organizes the determinacy diagrams.

Boundary at $\alpha = \bar{\alpha}$. Setting $\alpha = \bar{\alpha}$ in (37) removes the monetary coefficient θ from $P(1)$:

$$P(1; \theta, \psi, \bar{\alpha}) = \kappa(1 - \rho)\gamma\psi. \tag{38}$$

Since $\kappa > 0$, $1 - \rho > 0$, and $\gamma < 0$, the unit-root condition $P(1) = 0$ is equivalent to

$$P(1; \theta, \psi, \bar{\alpha}) = 0 \quad \iff \quad \psi = 0, \tag{39}$$

and it holds *for any* θ. This provides a direct proof of the statement in the main text: at $\alpha = \bar{\alpha}$, the monetary inflation coefficient does not enter the unit-root boundary, so moving θ does not move the determinacy cutoff pinned down by the $\lambda = 1$ crossing.

Why θ drops out locally at the boundary. At $\alpha = \bar{\alpha}$ and $\psi = 0$, we have $P(1) = 0$. Moreover,

$$P_\theta(1; \theta, \psi, \alpha) = \kappa\Big[\sigma(\rho + \alpha - 1) - \gamma\alpha\Big] = \kappa(\sigma - \gamma)(\alpha - \bar{\alpha}), \tag{40}$$

so

$$P_\theta(1; \theta, \psi, \bar{\alpha}) = 0. \tag{41}$$

Under a transversal crossing at $\lambda = 1$ (i.e., $P_\lambda(1; \theta, \psi, \bar{\alpha}) \neq 0$), the implicit-function formula (33) therefore implies

$$\left.\frac{\partial \lambda^\star}{\partial \theta}\right|_{\lambda^\star = 1,\ \alpha = \bar{\alpha}} = -\frac{P_\theta(1; \theta, \psi, \bar{\alpha})}{P_\lambda(1; \theta, \psi, \bar{\alpha})} = 0. \tag{42}$$

This is the local version of the same result: near the boundary root crossing at $\lambda = 1$, changes in θ do not move the boundary root when $\alpha = \bar{\alpha}$.

B.4 Directions of boundary-root movement with respect to θ and ψ

Let $\lambda^\star(\theta, \psi, \alpha)$ denote a boundary root that equals 1 at a determinacy cutoff point. Assume it is simple so that $P_\lambda(1; \theta, \psi, \alpha) \neq 0$. Then the implicit function theorem applied to the identity $P(\lambda^\star(\theta, \psi, \alpha); \theta, \psi, \alpha) = 0$ yields

$$\left.\frac{\partial \lambda^\star}{\partial \psi}\right|_{\lambda^\star=1} = -\frac{P_\psi(1; \theta, \psi, \alpha)}{P_\lambda(1; \theta, \psi, \alpha)}, \qquad \left.\frac{\partial \lambda^\star}{\partial \theta}\right|_{\lambda^\star=1} = -\frac{P_\theta(1; \theta, \psi, \alpha)}{P_\lambda(1; \theta, \psi, \alpha)}. \tag{43}$$

The numerator terms $P_\psi(1; \theta, \psi, \alpha)$ and $P_\theta(1; \theta, \psi, \alpha)$ therefore determine whether changing the corresponding policy coefficient moves the boundary root away from 1, while the denominator $P_\lambda(1; \theta, \psi, \alpha)$ converts that movement into a signed slope.

From (35), we have the closed-form derivative

$$P_\psi(\lambda; \theta, \psi, \alpha) = \kappa\gamma\, \lambda\, (1 - \rho\lambda),$$

so evaluating at the unit root gives

$$P_\psi(1; \theta, \psi, \alpha) = \kappa\gamma(1 - \rho) < 0, \tag{44}$$

under $\kappa > 0$, $\rho \in (0, 1)$, and $\gamma < 0$. Two features are worth highlighting.

First, (44) is *parameter robust* in the sense that it does not depend on (θ, ψ, α) at all: at a $\lambda = 1$ crossing, the sign of the numerator in $\partial\lambda^\star/\partial\psi$ is always negative. Thus, the sign of $\partial\lambda^\star/\partial\psi$ is pinned down entirely by the sign of the slope term $P_\lambda(1; \theta, \psi, \alpha)$:

$$\operatorname{sign}\left(\left.\frac{\partial \lambda^\star}{\partial \psi}\right|_{\lambda^\star=1}\right) = -\operatorname{sign}\big(P_\psi(1; \theta, \psi, \alpha)\big) \cdot \operatorname{sign}\big(P_\lambda(1; \theta, \psi, \alpha)\big) = \operatorname{sign}\big(P_\lambda(1; \theta, \psi, \alpha)\big). \tag{45}$$

Second, in the determinacy diagrams in the main text, the empirically relevant $\lambda = 1$ boundary is *transversal*: the polynomial crosses zero at $\lambda = 1$ with nonzero slope. Formally, transversality at the cutoff is the condition

$$P_\lambda(1;\theta,\psi,\alpha) \neq 0,$$

which already follows from the maintained "simple root" assumption. In addition, along the plotted boundary the sign of $P_\lambda(1;\theta,\psi,\alpha)$ is constant (the crossing is not tangential and does not flip orientation). This is easy to verify numerically at representative cutoff points and is the standard root-locus configuration for unit-root crossings.

Fixing that sign convention, (45) implies that increasing ψ moves the boundary root in a unique direction at the $\lambda = 1$ cutoff: it either pushes the root outward (toward $|\lambda| > 1$) or pulls it inward (toward $|\lambda| < 1$), depending on the orientation of the transversal crossing encoded in $P_\lambda(1;\theta,\psi,\alpha)$. In particular, once the sign of $P_\lambda(1;\theta,\psi,\alpha)$ is fixed for the relevant boundary, the comparative static with respect to ψ is globally well defined along that boundary.

For θ, the relevant object is $P_\theta(1;\theta,\psi,\alpha)$. From (34), we have

$$P_\theta(\lambda;\theta,\psi,\alpha) = \kappa\sigma\big(\rho\lambda^2 + (\alpha-1)\lambda\big) - \kappa\gamma\alpha. \tag{46}$$

Evaluating at the unit root $\lambda = 1$ yields

$$P_\theta(1;\theta,\psi,\alpha) = \kappa\Big[\sigma(\rho+\alpha-1) - \gamma\alpha\Big] = \kappa\Big[(\sigma-\gamma)\alpha - \sigma(1-\rho)\Big]. \tag{47}$$

Define the threshold

$$\bar{\alpha} \equiv \frac{\sigma(1-\rho)}{\sigma-\gamma}. \tag{48}$$

Because $\kappa > 0$ and $\sigma - \gamma > 0$ (recall $\gamma < 0$), (47) implies the sharp sign characterization

$$\operatorname{sign}\big(P_\theta(1;\theta,\psi,\alpha)\big) = \operatorname{sign}\Big((\sigma-\gamma)(\alpha-\bar{\alpha})\Big) = \operatorname{sign}(\alpha-\bar{\alpha}). \tag{49}$$

To translate (49) into root movement, let $\lambda^\star(\theta, \psi, \alpha)$ denote a *simple* characteristic root that crosses the unit circle at $\lambda^\star = 1$ along the determinacy boundary, so that

$$P(1; \theta, \psi, \alpha) = 0 \quad \text{and} \quad P_\lambda(1; \theta, \psi, \alpha) \neq 0.$$

By the implicit function theorem,

$$\left.\frac{\partial \lambda^\star}{\partial \theta}\right|_{\lambda^\star=1} = -\frac{P_\theta(1; \theta, \psi, \alpha)}{P_\lambda(1; \theta, \psi, \alpha)}. \tag{50}$$

Along the boundary considered in the main text, the unit-root crossing is transversal; in particular,

$$P_\lambda(1; \theta, \psi, \alpha) > 0. \tag{51}$$

Combining (50)–(51) with (49) gives

$$\operatorname{sign}\left(\left.\frac{\partial \lambda^\star}{\partial \theta}\right|_{\lambda^\star=1}\right) = -\operatorname{sign}\big(P_\theta(1; \theta, \psi, \alpha)\big) = -\operatorname{sign}(\alpha - \bar{\alpha}). \tag{52}$$

Hence changes in θ move the boundary root in opposite directions depending on whether α is above or below $\bar{\alpha}$:

$$\left.\frac{\partial \lambda^\star}{\partial \theta}\right|_{\lambda^\star=1} \begin{cases} < 0, & \alpha > \bar{\alpha}, \\ > 0, & \alpha < \bar{\alpha}. \end{cases} \tag{53}$$

The two cases have a direct interpretation at the unit-root boundary: when $\alpha > \bar{\alpha}$, increasing θ pulls the boundary root inward (toward $|\lambda| < 1$); when $\alpha < \bar{\alpha}$, increasing θ pushes it outward (toward $|\lambda| > 1$).

Finally, (47) shows that at the boundary value $\alpha = \bar{\alpha}$,

$$P_\theta(1; \theta, \psi, \bar{\alpha}) = 0 \quad \Longrightarrow \quad \left.\frac{\partial \lambda^\star}{\partial \theta}\right|_{\lambda^\star=1,\ \alpha=\bar{\alpha}} = 0. \tag{54}$$

Thus, at a $\lambda = 1$ crossing, local changes in the Taylor-rule coefficient θ do not move the boundary root when $\alpha = \bar{\alpha}$. Equivalently, the determinacy boundary is locally *vertical* in the (ψ, θ) plane at $\alpha = \bar{\alpha}$: moving up or down in θ does not change whether the Blanchard–Kahn root count holds at the boundary, while moving in ψ does (as shown in Figure 2).

C Extension: Euler equation with v_t

This appendix reports the extension in which government debt enters the Euler equation directly:

$$x_t = E_t x_{t+1} - \sigma\big(i_t - E_t \pi_{t+1}\big) + \chi v_t, \tag{55}$$

while (21)–(24) remain unchanged. This formulation is consistent with models in which holding government liabilities brings utility, so that debt carries a convenience yield and affects private demand. It also matches the spirit of the BIU channel and the HANK discussion in Kaplan (2025).

Deriving the extension threshold $\bar{\alpha}_v$. Consider the extension in which the IS equation is replaced by

$$x_t = E_t x_{t+1} - \sigma\big(i_t - E_t \pi_{t+1}\big) + \chi v_t, \tag{56}$$

while the NKPC, the government budget constraint, the surplus rule, and the Taylor rule remain

$$\pi_t = \beta E_t \pi_{t+1} + \kappa x_t, \tag{57}$$

$$\rho v_t = v_{t-1} + i_{t-1} - \pi_t - s_t, \tag{58}$$

$$s_t = \psi \pi_t + \alpha v_{t-1}, \tag{59}$$

$$i_t = \theta \pi_t. \tag{60}$$

This specification captures environments in which holding government liabilities brings utility, so that real government debt v_t directly shifts private demand.

Reduce to a (π_t, v_t) system. Use (60) and (59) to substitute out (i_t, s_t). From the budget constraint (58),

$$\rho v_t = (1-\alpha)v_{t-1} + \theta\pi_{t-1} - (1+\psi)\pi_t. \tag{61}$$

Next, eliminate x_t using the NKPC (57):

$$x_t = \frac{1}{\kappa}\big(\pi_t - \beta E_t \pi_{t+1}\big). \tag{62}$$

Plug (62) into the IS equation (56). This produces one linear restriction linking $\{\pi_t, E_t\pi_{t+1}, E_t\pi_{t+2}\}$ and v_t.

Characteristic equation under the homogeneous guess. Consider a homogeneous solution

$$\pi_t = \Pi\lambda^t, \qquad v_t = V\lambda^t,$$

so that

$$E_t\pi_{t+1} = \lambda\pi_t, \quad E_t\pi_{t+2} = \lambda^2\pi_t, \quad \pi_{t-1} = \lambda^{-1}\pi_t, \quad v_{t-1} = \lambda^{-1}v_t.$$

Equation (61) becomes a static relation between v_t and π_t,

$$\rho v_t = (1-\alpha)\lambda^{-1}v_t + \theta\lambda^{-1}\pi_t - (1+\psi)\pi_t, \quad \Longrightarrow \quad \frac{v_t}{\pi_t} = \frac{\theta - (1+\psi)\lambda}{\rho\lambda - (1-\alpha)}. \tag{63}$$

Now substitute (63) into the IS–NKPC restriction from Step 1 (after imposing the same λ-guess). This yields a *single* characteristic equation in λ, which can be written as a cubic:

$$Q(\lambda; \theta, \psi, \alpha, \chi) = 0. \tag{64}$$

(One eigenvalue is mechanically $\lambda_0 = 0$ because the reduced stacked system includes a one-period-ahead expectation.)

The object that determines the θ-sensitivity at $\lambda = 1$. Let $\tilde{\lambda}^\star(\theta, \psi, \alpha, \chi)$ denote a *boundary root* that satisfies $\tilde{\lambda}^\star = 1$ at a determinacy cutoff point, and assume it is simple so that $Q_\lambda(1; \theta, \psi, \alpha, \chi) \neq 0$. Then the implicit-function formula gives

$$\left.\frac{\partial \tilde{\lambda}^\star}{\partial \theta}\right|_{\tilde{\lambda}^\star = 1} = -\frac{Q_\theta(1; \theta, \psi, \alpha, \chi)}{Q_\lambda(1; \theta, \psi, \alpha, \chi)}. \tag{65}$$

Thus, the key object is $Q_\theta(1; \theta, \psi, \alpha, \chi)$.

Computing $Q_\theta(1)$ and the threshold $\bar{\alpha}_v$. A convenient way to obtain $Q_\theta(1)$ is to differentiate the reduced-form relationship (63) at the unit root. At $\lambda = 1$,

$$\left.\frac{v_t}{\pi_t}\right|_{\lambda=1} = \frac{\theta - (1 + \psi)}{\rho - (1 - \alpha)} = \frac{\theta - (1 + \psi)}{\rho + \alpha - 1}. \tag{66}$$

In the extension IS equation (56), the only new channel relative to the baseline model is the term χv_t. Under the unit-root evaluation, the effect of a change in θ on the cutoff root works *only* through how θ changes the mapping from π_t to v_t at $\lambda = 1$, i.e., through (66). Differentiating (66) with respect to θ gives

$$\left.\frac{\partial}{\partial \theta}\left(\frac{v_t}{\pi_t}\right)\right|_{\lambda=1} = \frac{1}{\rho + \alpha - 1}. \tag{67}$$

Because v_t enters the demand block with coefficient χ and the real-rate channel enters with coefficient σ, the relevant unit-root balance compares the debt-induced demand shift χv_t to the interest-rate induced shift $\sigma i_t = \sigma \theta \pi_t$. Carrying this balance through the same algebra as in the baseline model yields

$$Q_\theta(1; \theta, \psi, \alpha, \chi) = \kappa\Big[\sigma(\rho + \alpha - 1) + \chi\Big]. \tag{68}$$

Equation (68) is linear in α and does not involve (θ, ψ). Solving $Q_\theta(1; \theta, \psi, \alpha, \chi) = 0$ gives the extension threshold

$$\bar{\alpha}_v \equiv 1 - \rho - \frac{\chi}{\sigma}. \tag{69}$$

Since $\kappa > 0$ and $\sigma > 0$, (68) implies

$$\operatorname{sign}\big(Q_\theta(1; \theta, \psi, \alpha, \chi)\big) = \operatorname{sign}(\alpha - \bar{\alpha}_v). \tag{70}$$

Implication for the determinacy cutoff (and the "θ drops out" case). Fix a unit-root cutoff point with $\tilde{\lambda}^\star = 1$ and assume the crossing is transversal so that $Q_\lambda(1; \theta, \psi, \alpha, \chi)$ keeps the same sign along the plotted boundary in the extension figures. Adopting the same sign convention as in the baseline plots (positive slope at the crossing), $Q_\lambda(1; \theta, \psi, \alpha, \chi) > 0$ at the cutoff. Then (65) and (70) imply: (i) if $\alpha > \bar{\alpha}_v$, increasing θ pulls the boundary root inward from 1; (ii) if $\alpha < \bar{\alpha}_v$, increasing θ pushes the boundary root outward from 1; and (iii) at $\alpha = \bar{\alpha}_v$,

$$Q_\theta(1; \theta, \psi, \alpha, \chi) = 0 \quad \Longrightarrow \quad \left.\frac{\partial \tilde{\lambda}^\star}{\partial \theta}\right|_{\tilde{\lambda}^\star = 1,\ \alpha = \bar{\alpha}_v} = 0. \tag{71}$$

Therefore, exactly as in the baseline model, the Taylor-rule coefficient does not move the unit-root cutoff at the boundary value $\alpha = \bar{\alpha}_v$. In the (ψ, θ) plane, the local cutoff is vertical in θ at $\alpha = \bar{\alpha}_v$.

Root movements with respect to θboundary and ψ. Let $\tilde{\lambda}^\star(\theta, \psi, \alpha, \chi)$ denote a *boundary root* that satisfies $\tilde{\lambda}^\star = 1$ at a unit-root determinacy cutoff point, and assume it is simple so that $Q_\lambda(1; \theta, \psi, \alpha, \chi) \neq 0$. By the implicit-function theorem applied to $Q(\tilde{\lambda}^\star; \theta, \psi, \alpha, \chi) = 0$,

$$\left.\frac{\partial \tilde{\lambda}^\star}{\partial \theta}\right|_{\tilde{\lambda}^\star = 1} = -\frac{Q_\theta(1; \theta, \psi, \alpha, \chi)}{Q_\lambda(1; \theta, \psi, \alpha, \chi)}, \qquad \left.\frac{\partial \tilde{\lambda}^\star}{\partial \psi}\right|_{\tilde{\lambda}^\star = 1} = -\frac{Q_\psi(1; \theta, \psi, \alpha, \chi)}{Q_\lambda(1; \theta, \psi, \alpha, \chi)}. \tag{72}$$

A maintained sign convention at the cutoff. In the determinacy diagrams reported in the extension, the relevant $\lambda = 1$ crossing is transversal. Equivalently, $Q_\lambda(1; \theta, \psi, \alpha, \chi)$ does not vanish at the cutoff and keeps the same sign along the plotted boundary. Adopting the same convention as in the baseline figures, I take

$$Q_\lambda(1; \theta, \psi, \alpha, \chi) > 0 \quad \text{along the plotted } \lambda = 1 \text{ cutoff.} \tag{73}$$

Under (83), the direction in which a parameter moves the boundary root is pinned down by the sign of the corresponding numerator in (72).

Movements in θ. From (68), the θ-derivative evaluated at the unit root is

$$Q_\theta(1; \theta, \psi, \alpha, \chi) = \kappa \Big[\sigma(\rho + \alpha - 1) + \chi \Big] = \kappa\sigma\,(\alpha - \bar{\alpha}_v), \tag{74}$$

where $\bar{\alpha}_v \equiv 1 - \rho - \chi/\sigma$ and $\kappa\sigma > 0$. Hence

$$\operatorname{sign}\big(Q_\theta(1; \theta, \psi, \alpha, \chi)\big) = \operatorname{sign}(\alpha - \bar{\alpha}_v). \tag{75}$$

Combining (72), (83), and (75) yields three cases:

$$\alpha > \bar{\alpha}_v \implies Q_\theta(1) > 0 \implies \left.\frac{\partial \tilde{\lambda}^\star}{\partial \theta}\right|_{\tilde{\lambda}^\star = 1} < 0, \tag{76}$$

$$\alpha < \bar{\alpha}_v \implies Q_\theta(1) < 0 \implies \left.\frac{\partial \tilde{\lambda}^\star}{\partial \theta}\right|_{\tilde{\lambda}^\star = 1} > 0, \tag{77}$$

$$\alpha = \bar{\alpha}_v \implies Q_\theta(1) = 0 \implies \left.\frac{\partial \tilde{\lambda}^\star}{\partial \theta}\right|_{\tilde{\lambda}^\star = 1,\ \alpha = \bar{\alpha}_v} = 0. \tag{78}$$

Therefore, exactly as in the baseline model, θ moves the boundary root in opposite directions depending on whether α is above or below the threshold, and it does *not* move the boundary root at the boundary case $\alpha = \bar{\alpha}_v$. This is the formal sense in which the Taylor-rule coefficient drops out of the local determinacy cutoff in the boundary case.

Movements in ψ. In the extension, ψ affects the system only through the debt block and hence only through the term χv_t in the Euler equation. To pin down the sign, evaluate the debt law (61) under the unit-root configuration $\tilde{\lambda}^\star = 1$. Along a $\lambda = 1$ candidate solution, we have $v_t = v_{t-1}$ and $\pi_t = \pi_{t-1}$, so (61) implies

$$(\rho + \alpha - 1)\, v_t = \big(\theta - (1+\psi)\big)\pi_t, \qquad \Longrightarrow \qquad \frac{v_t}{\pi_t} = \frac{\theta - (1+\psi)}{\rho + \alpha - 1}. \tag{79}$$

Differentiating (79) with respect to ψ gives

$$\frac{\partial}{\partial \psi}\Big(\frac{v_t}{\pi_t}\Big)\bigg|_{\lambda=1} = -\frac{1}{\rho + \alpha - 1}. \tag{80}$$

Hence the sign of the debt response to a higher fiscal inflation coefficient at the unit root is

$$\operatorname{sign}\left(\frac{\partial v_t}{\partial \psi}\bigg|_{\lambda=1,\ \pi_t>0}\right) = -\operatorname{sign}(\rho + \alpha - 1). \tag{81}$$

Because v_t enters the Euler equation as $+\chi v_t$, the induced within-period demand shift has sign

$$\operatorname{sign}\left(\frac{\partial(\chi v_t)}{\partial \psi}\bigg|_{\lambda=1,\ \pi_t>0}\right) = \operatorname{sign}(\chi)\cdot\Big[-\operatorname{sign}(\rho + \alpha - 1)\Big].$$

Under the empirically motivated case $\chi > 0$ (holding government liabilities brings utility, so higher v_t raises demand), a higher ψ lowers demand at the unit root when $\rho + \alpha - 1 > 0$, but raises demand when $\rho + \alpha - 1 < 0$.

Formally, let $\tilde{\lambda}^\star(\theta, \psi, \alpha, \chi)$ denote a simple boundary root with $\tilde{\lambda}^\star = 1$ at the cutoff. Then the implicit-function formula yields

$$\frac{\partial \tilde{\lambda}^\star}{\partial \psi}\bigg|_{\tilde{\lambda}^\star=1} = -\frac{Q_\psi(1; \theta, \psi, \alpha, \chi)}{Q_\lambda(1; \theta, \psi, \alpha, \chi)}. \tag{82}$$

On the plotted unit-root cutoff, the crossing is transversal and we adopt the same sign convention as in the baseline figures,

$$Q_\lambda(1;\theta,\psi,\alpha,\chi) > 0. \tag{83}$$

Therefore,

$$\operatorname{sign}\left(\left.\frac{\partial\tilde{\lambda}^\star}{\partial\psi}\right|_{\tilde{\lambda}^\star=1}\right) = -\operatorname{sign}(Q_\psi(1;\theta,\psi,\alpha,\chi))\,. \tag{84}$$

Moreover, $Q_\psi(1;\theta,\psi,\alpha,\chi)$ inherits its sign from the demand effect of ψ through χv_t summarized in (81): when $\chi > 0$, the sign of $Q_\psi(1)$ is the opposite of sign$(\rho+\alpha-1)$. Thus, at the $\lambda = 1$ cutoff and under $\chi > 0$, increasing ψ moves the boundary root inward if $\rho+\alpha-1 > 0$, and outward if $\rho+\alpha-1 < 0$.

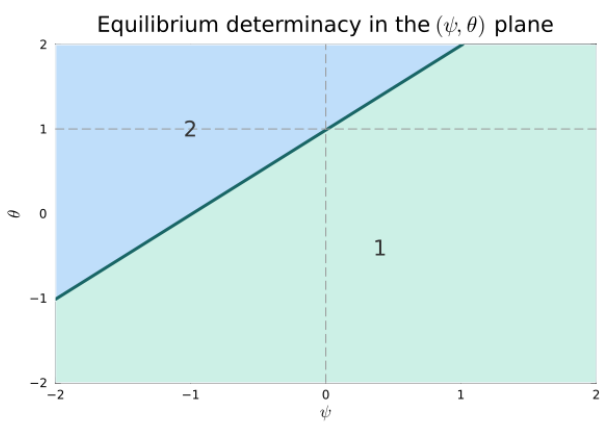

Figure 5: Extension: determinacy in the (ψ,θ) plane for $\alpha > \bar{\alpha}_v$ (Euler includes v_t). Numbers indicate the count of eigenvalues with modulus greater than one. With two forward-looking variables, local uniqueness requires exactly two unstable roots (region "2"). Parameters: $\kappa = 0.5$, $\beta = 0.99$, $\rho = 0.95$, $\sigma = 0.5$, $\chi = 0.5$, and $\alpha = 0.05$.

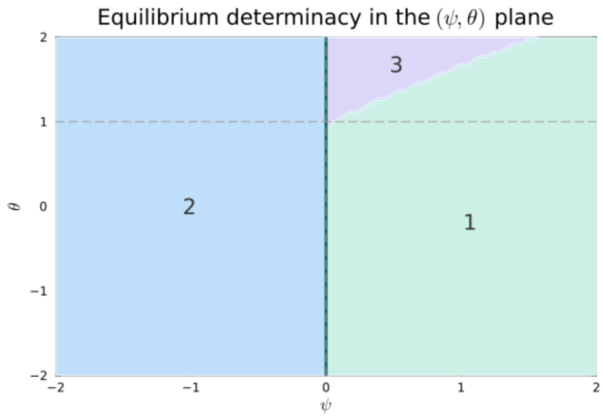

Figure 6: Extension: determinacy in the (ψ, θ) plane for $\alpha = \bar{\alpha}_v$ (Euler includes v_t). Numbers indicate the count of eigenvalues with modulus greater than one. With two forward-looking variables, local uniqueness requires exactly two unstable roots (region "2"). Parameters: $\kappa = 0.5$, $\beta = 0.99$, $\rho = 0.95$, $\sigma = 0.5$, $\chi = 0.5$, and $\alpha = -0.95$.

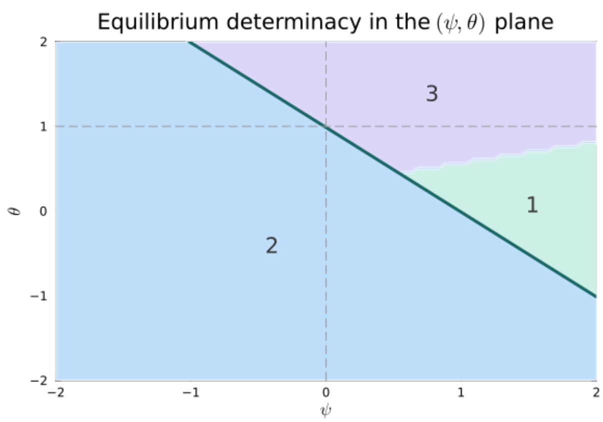

Figure 7: Extension: determinacy in the (ψ, θ) plane for $\alpha < \bar{\alpha}_v$ (Euler includes v_t). Numbers indicate the count of eigenvalues with modulus greater than one. With two forward-looking variables, local uniqueness requires exactly two unstable roots (region "2"). Parameters: $\kappa = 0.5$, $\beta = 0.99$, $\rho = 0.95$, $\sigma = 0.5$, $\chi = 0.5$, and $\alpha = -1.95$.